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E1. Section 6.2.3 (p.16, l.1346-1355) contains a duplicated paragraph ("Certain external shocks ... carbon emissions performance." appears twice). Please delete the duplicate so the paragraph appears only once.
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E4. Section 3.4 (p.9, l.541): "establishing the baseline baseline effect" -> "establishing the baseline effect".
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E6. Section 4.1 (p.10, l.588): "the primary explanatory variable of interest., The" -> remove the extra comma ("of interest. The").
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The impact of green funds on the carbon emission intensity of listed companies

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This paper investigates the impact of green funds on corporate carbon emission intensity using data from China's A-share listed firms. The empirical results indicate that green funds significantly reduce corporate carbon emission intensity, a finding that remains robust after a series of alternative measures and endogeneity treatments. Heterogeneity analysis reveals that this emission-reduction effect is more pronounced among firms in non-heavily polluting industries, regions with stringent environmental regulations, growth-stage enterprises, and industries with low concentration. Mechanism analysis suggests that green funds curb carbon emission intensity primarily by promoting green innovation, alleviating financing constraints, fulfilling ESG responsibilities, enhancing corporate reputation, and driving biased technological progress. Furthermore, digital technologies, managers' educational backgrounds, and the adoption of artificial intelligence play positive moderating roles. This study provides empirical evidence on how green financial instruments drive micro-level low-carbon transitions and offers policy insights for optimizing green fund allocations.

KEYWORDS
carbon emission intensity, digital technologies, ESG, green funds, green innovation

1 Introduction

As the global climate change crisis intensifies, advancing low-carbon transition has become a core strategy for nations pursuing sustainable development. Against the backdrop of its dual carbon goals, China is progressively shifting from high-speed growth towards a high-quality, green, and low-carbon development model, accompanied by the introduction of a series of green policies and institutional arrangements. As a vital policy instrument guiding optimal resource allocation and supporting green economic transformation, green finance is increasingly emerging as a key driver in promoting corporate pollution reduction and carbon mitigation alongside fostering green technological innovation. It has consequently become a prominent research focus within academic circles in recent years.

As early as 2011, the State Council Information Office of the People's Republic of China issued the "China's Policies and Actions on Climate Change (2011)", which stated that the "Twelfth Five-Year Plan for National Economic and Social Development of the People's Republic of China", formulated and implemented in 2011, established the policy orientation for green and low-carbon development over the next 5 years and clarified the objectives and tasks for addressing climate change. In 2015, the Chinese government formally released the "Overall Plan for Reforming the Ecological Civilisation System", explicitly proposing the establishment of a green financial system. This initiative aimed to advance market mechanisms for environmental governance and ecological conservation, utilise various

financial instruments to drive structural transformation in capital markets, and support enterprises in achieving pollution reduction and carbon emission targets. Existing research has demonstrated that green financial instruments play a positive role in enhancing corporate environmental performance. For instance, green credit (Lee et al., 2022; Sun and Zeng, 2023; Tan et al., 2025; Yan and Gong, 2024), green bonds (Zhang and Guo, 2025; ~~Lau, 2025~~), and green funds (Li et al., 2025; Wang and Jin, 2025) have all contributed to improving corporate Carbon Emission Intensity to varying degrees.

Among the array of green financial instruments, green funds, as equity-based capital tools, exhibit significant differences from green loans and green bonds in terms of governance mechanisms, incentive models, capital allocation pathways, and risk preferences. Consequently, they demonstrate unique advantages in supporting the green transition of the real economy. Existing research indicates that green funds primarily drive improvements in corporate carbon emissions by alleviating resource and financing constraints faced by green enterprises (Li et al., 2025) and enhancing managers' environmental awareness and responsibility (Wang and Jin, 2025). However, systematic research remains scarce on whether green funds can further influence corporate Carbon Emission Intensity through alternative pathways—such as technological incentives, public oversight, or corporate reputation building. Particularly at the micro-firm level, the identification effects and transmission mechanisms warrant deeper exploration.

Concurrently, with the widespread application of artificial intelligence (AI) and digital technologies, some studies suggest these can generate synergistic effects with green finance, amplifying corporate environmental performance (~~Peng et al., 2024;~~ Xu et al., 2024). Against this backdrop, examining whether the environmental effects of green funds are amplified under digital conditions and how the underlying mechanisms evolve has become a significant topic warranting attention. Furthermore, the impact of green funds on corporate carbon emissions may not be uniform. Environmental performance may exhibit pronounced heterogeneity, influenced by multiple factors including the industry sector, regional institutional environment, life cycle stage, and level of financial development.

Therefore, systematically investigating how and under what circumstances green funds effectively influence corporate Carbon Emission Intensity not only contributes to expanding the theoretical boundaries of green finance research but also holds significant theoretical value and practical relevance for refining green investment governance mechanisms and advancing the strategic objectives of carbon peaking and carbon neutrality.

Existing research closely related to the present study's theme includes Li et al. (2025) and Wang and Jin (2025). Li et al. examined the impact of government green funds on corporate carbon emissions, finding they primarily exert influence through pathways such as resource allocation optimisation, innovation incentives, and market structure adjustments. Wang and Jin, meanwhile, focused on how green funds guide low-carbon corporate behaviour by enhancing managers' (particularly CEOs') environmental awareness. By contrast, this paper builds upon existing research by introducing the corporate reputation pathway as a novel mechanism through which green funds influence corporate carbon emissions. It further explores the potential moderating role of digital technologies and AI within this framework, aiming to systematically delineate the operational mechanisms and conditional boundaries of green funds' impact on corporate Carbon Emission Intensity from a multidimensional perspective.

The principal marginal contributions of this paper are articulated as follows:

First, this paper deepens the mechanism framework of green funds by uncovering a "dual-incentive" path of corporate reputation and biased technological progress, moving beyond the traditional "resource allocation" and "environmental awareness" perspectives emphasized by ~~Li et al.~~ While previous literature largely treats corporate green behavior as a passive response to fund injection, we introduce corporate reputation as an external market discipline and internal incentive. By matching this with biased technological progress, this study systematically illustrates how green funds transform external environmental pressure into autonomous, long-term low-carbon strategies, thereby offering a more comprehensive theoretical mechanism for corporate emission reductions.

Second, rather than treating digital technologies and artificial intelligence (AI) as isolated control factors, this study explores their compounding and amplifying effects when coupled with green funds, bridging a critical gap in the research by Wang and Jin. Unlike existing green finance literature that examines financial tools and digital intelligence separately, we examine how digital technologies and AI act as boundary conditions—functioning as "catalysts" and "amplifiers" for green funds. We demonstrate that digital integration helps mitigate the green fund's inherent informational asymmetry in tracking carbon data, while AI optimizes energy efficiency at the micro-level, providing fresh, contextual insights into the compounding environmental governance effects of "Finance + Tech."

Third, this paper delineates the precise conditional boundaries of green funds across multi-dimensional macro-micro linkages. By comprehensively interlocking corporate industry attributes, regional environmental regulations, and corporate life-cycle stages, this study maps out a granular heterogeneity matrix. This addresses the overly generalized conclusions in previous literature, clarifying the exact scenarios where green fund policy instruments achieve maximum precision and efficiency, and offering highly differentiated governance recommendations for policymakers.

2 Literature review

2.1 Research on carbon emission intensity in industrial enterprises

Against the backdrop of the dual carbon strategy, industrial enterprises—as primary sources of carbon emissions—have drawn significant attention from academia and policymakers regarding their emission reduction practices and performance. Carbon emissions, serving as a crucial metric for gauging environmental externalities within the industrial sector, have consistently formed a central focus in environmental economics and green development research.

Currently, numerous scholars employ diverse methodologies to evaluate corporate Carbon Emission Intensity. The first approach is

the carbon emission factor method. This technique directly calculates carbon emissions based on energy consumption data and carbon emission coefficients (Cui et al., 2021), subsequently enabling the computation of carbon emission intensity. This method finds extensive application in corporate-level assessments. The second approach encompasses Life Cycle Assessment (LCA) and Input-Output (I-O) analysis. The core principle of LCA lies in assessing all environmental burdens associated with a product or service, tracing from raw material extraction through to waste disposal (Klöpffer, 1997). However, while this method is relatively suitable for macro-level carbon accounting, its application at the enterprise level is constrained.

Existing research has examined factors influencing corporate carbon emissions from multiple dimensions. One such dimension concerns firm characteristics and governance structures. Firm size, ownership structure, capital intensity, and export status significantly impact carbon emission levels (Holladay, 2010; Cui et al., 2016). For instance, large enterprises often exhibit lower carbon intensity than small firms due to stronger institutional norms and external oversight. Furthermore, state-owned enterprises face greater social responsibility pressures and tend to allocate more resources towards environmental protection (Wang et al., 2022). Secondly, technological efficiency and green innovation. Technological advancement represents the fundamental pathway to reducing carbon emissions. Research indicates that improvements in total factor productivity significantly suppress carbon emission growth (Jin et al., 2017), while the quantity and intensity of green patents markedly reduce carbon intensity (Su and Moaniba, 2017). Thirdly, environmental regulation and policy incentives. Environmental regulations exert dual effects on corporate carbon emissions: a 'mandatory effect' and an 'incentive effect'. The Porter hypothesis has been partially validated in certain studies, indicating that moderate regulation can incentivise firms to pursue green innovation, thereby achieving a 'win-win' outcome for environmental protection and profits (Farooq et al., 2024).

2.2 Green finance and corporate carbon emissions performance

As global policies addressing climate change progressively tighten, the financial system's role in driving the low-carbon transition has become increasingly prominent. Broadly defined, green finance encompasses financial instruments and policy frameworks such as green credit, green bonds, green funds, green insurance, and green policies. It aims to channel resources towards sustainable sectors including environmental protection, energy conservation, and emissions reduction through financial means. Existing research indicates that green financial instruments can effectively enhance corporate environmental performance and reduce carbon emissions.

Green credit, as a vital component of green finance, has been demonstrated by multiple studies to play a positive role in constraining financing for highly polluting enterprises and directing resources towards green sectors (Lee et al., 2022; Sun and Zeng, 2023; Yan and Gong, 2024). Green bonds are also extensively utilised to support investments in low-carbon technology projects. Flammer (2021) found that green bond issuance not only enhances corporate environmental reputations but also stimulates financial backing and implementation of green projects (Zhang and Guo, 2025; Lau, 2025). Furthermore, government-established green funds provide stable financing for early-stage technological R&D and green industry cultivation, thereby underpinning corporate green transitions (Li et al., 2025).

Green funds, as a form of capital organisation guided by environmental, social and governance (ESG) investment criteria, are regarded as placing greater emphasis on the synergistic development of long-term financial and environmental performance within enterprises (Martin and Moser, 2016). They typically adhere to the Principles for Responsible Investment (PRI) and undergo scrutiny and evaluation from the public, investors, and professional third parties, thereby imposing heightened environmental and sustainability requirements on investee companies (Majoch et al., 2017).

2.3 Research into the mechanisms by which green funds influence corporate carbon emissions performance

Green funds influence corporate carbon emissions behaviour through multiple channels, with existing literature primarily exploring mechanisms concerning resource allocation, innovation incentives, industrial restructuring, and corporate governance. Firstly, they enhance resource allocation efficiency. As a market-oriented capital mechanism, green funds help optimise resource allocation, particularly by providing crucial financial support to small and medium-sized enterprises (SMEs) unable to undertake effective green transformation due to financing constraints (Guo and Fang, 2024; Shao et al., 2023; Chi et al., 2023). Li et al. (2025) note that green fund investments can mitigate resource misallocation by redirecting capital towards low-carbon, high-efficiency enterprises. Secondly, they elevate the capital costs for polluting firms. By excluding polluting enterprises from investment portfolios and intensifying financial scrutiny on high-emission companies, green funds indirectly raise their financing costs, thereby compelling environmental remediation or transformation (Engle et al., 2020). Thirdly, they incentivise green innovation. By allocating capital to support low-carbon technology R&D, green funds alleviate enterprises' financial constraints in developing green technologies, thereby promoting the adoption of low-carbon products and processes (Li et al., 2024; Guo et al., 2024; Tan et al., 2025). Government-established green guiding funds often catalyse greater private capital investment in green innovation projects, forming a 'fiscal + market' green synergy model (Lerner, 2020). Fourthly, they facilitate industrial structure optimisation and corporate green transformation. Hu et al. (2020) indicate that green finance policies drive industrial upgrading towards green and low-carbon directions. Kuzmina et al. (2023) found that enterprises receiving green investment support are more likely to achieve green transformation and expand their operational scale, thereby transitioning from high-carbon to low-carbon production methods. Fifthly, it enhances corporate governance standards and environmental awareness. Relevant research indicates that green investment can influence strategic decision-making by strengthening CEOs' recognition of environmental responsibilities (Liu et al., 2023; Zhu et al., 2025). Moreover,

individual characteristics such as a CEO's educational background and environmental awareness may exert moderating effects on the relationship between green funds and Carbon Emission Intensity (Kang et al., 2025; Xu and Shi, 2025). Sixthly, ESG performance and capital market pressures. Research by Liu et al. and Wang et al. (2024) indicates growing capital market scrutiny of corporate environmental performance. Investor preference for enterprises with green development potential enables companies with strong ESG performance to secure lower-cost financing (Dahiya and Singh, 2021), further incentivising enhanced carbon emission governance. Seventh is public oversight and enhanced green discourse authority. Green funds typically sign PRI agreements and must disclose investment performance and environmental impacts to the public, subjecting themselves to external scrutiny. Majoch et al. (2017) emphasise that green funds often serve as "pioneers" in fulfilling green investment commitments, propagating green principles throughout capital markets and amplifying the "social echo" of green governance.

2.4 Literature summary and review

While existing literature provides a solid foundation for understanding the environmental effects of green finance, several critical research gaps remain unaddressed, failing to fully capture the distinctive nature of green funds and the shifting landscape of corporate low-carbon transitions.

First, prior studies often treat green finance as a homogeneous concept, largely failing to articulate why green funds are fundamentally different from green credit and green bonds in driving emission reductions. Unlike green credit and green bonds, which are debt-financing instruments characterized by rigid repayment terms and a primary focus on short-to-medium-term risk control, green funds function as equity-based capital with a longer investment horizon ("patient capital"). Consequently, green funds do not merely act as passive resource allocators; instead, they embed deep corporate governance mechanisms, directly aligning the investors' long-term sustainability goals with the investee firms' green strategies.

Second, the mechanism framework in current research exhibits a notable gap regarding the interactive nature of external market pressures and internal behavioral adjustments. Existing studies predominantly analyze direct financial channels (e.g., funding costs), while overlooking the role of corporate reputation and ESG performance as critical transmission mechanisms. In an era of heightened capital market scrutiny, green funds—bound by the Principles for Responsible Investment (PRI)—propagate strict environmental standards. This makes corporate reputation an essential conduit through which external market discipline is converted into an internal proactive incentive, while ESG performance acts as the standardized signaling mechanism that validates these improvements to the public.

Third, existing analytical frameworks frequently treat the financial environment and the technological landscape as independent domains, creating an artificial separation between green finance and digital intelligence. In reality, green funds face inherent informational asymmetry when tracking micro-level carbon data, which can lead to greenwashing. Introducing digital technology and artificial intelligence (AI) into the framework is therefore necessary, not as clichéd control factors, but as essential boundary conditions. Digitalization acts as an informational "catalyst" that reduces data transparency barriers, while AI functions as an operational "amplifier" that optimizes energy efficiency at the manufacturing level.

By addressing these omissions, this paper directly bridges the gaps in previous studies ~~(e.g., Li et al.; Wang and Jin)~~ by providing a more integrated "Finance + Tech" and "Governance + Reputation" framework to evaluate the true impact of green funds on corporate carbon emission intensity.

3 Theoretical analysis

3.1 Foundational layer: capital injection and baseline impact

With the ongoing advancement of the dual-carbon strategy, green finance plays an increasingly prominent role in driving corporate low-carbon transformation. As a vital component of the green finance system, green funds employ distinctive investment orientations and market discipline mechanisms to influence corporate Carbon Emission Intensity (CEI) through multiple pathways.

Existing literature indicates that an increase in green fund shareholdings strengthens the capital market's functions in resource allocation and corporate governance (Li et al., 2024; Tan et al., 2025). Although green funds employ strict environmental screening criteria, their core attributes remain those of investment funds. Consequently, their most fundamental function is providing "patient capital" to alleviate enterprises' financing constraints and barriers to resource acquisition. By injecting targeted capital, green funds expand the green production scale and optimize the energy consumption structure of recipient firms (Kuzmina et al., 2023). This direct financial support allows firms to absorb the high upfront costs of environmental compliance, shifting from defensive environmental practices to proactive carbon reduction strategies. Therefore, we propose the core baseline hypothesis:

Hypothesis 1. (H1): Green fund shareholdings have a significant negative impact on corporate carbon emission intensity.

3.2 Governance layer: reputational incentives and ESG market discipline

The mere provision of capital does not automatically yield emission reduction effects; it requires internal and external governance mechanisms to guide corporate intent.

From an internal governance perspective, the cognitive values of corporate leadership profoundly influence strategic resource allocation (Hambrick, 2007). Green funds, acting as institutional investors with strong environmental preferences, influence executive decision-making through pressure transmission and active governance participation (Zhu et al., 2025). This institutional presence prompts CEOs to adopt environmentally friendly production practices, implicitly enhancing corporate environmental responsibility awareness and driving firms to optimize carbon performance (Cheng et al., 2024).

From an external governance perspective, green funds indirectly promote carbon reductions by enhancing firms' Environmental, Social, and Governance (ESG) performance. A core mechanism driving this is the requirement for companies to align with international responsible investment frameworks, such as the UN PRI. Following this alignment, enterprises must regularly disclose detailed ESG metrics, subjecting themselves to independent third-party audits and public oversight (Guo and Bai, 2019). According to reputation theory (Barney, 1991), a favorable corporate ESG reputation acts as valuable intangible capital. To maintain this reputational capital and prevent capital divestment, firms face strong market discipline to fulfill environmental responsibilities. Consequently, we propose the following two mechanism hypotheses:

Hypothesis 2. (H2 / Mechanism 1): Green funds reduce corporate carbon emission intensity by enhancing the CEO's environmental awareness.

Hypothesis 3. (H3 / Mechanism 2): Green funds reduce corporate carbon emission intensity by improving corporate ESG performance and corporate reputation.

3.3 Execution layer: biased technological change and boundary conditions

At the terminal execution layer, capital and governance pressures must translate into technical realities. Acemoglu et al. (2012) note that under environmental constraints, market incentives can steer the direction of technological progress, encouraging firms to reduce reliance on polluting inputs and transition to clean inputs—a process known as biased technological change. Green funds function as a market-based quasi-subsidy that fuels green technological innovation, allowing firms to replace fossil energy with clean energy sources (Lin and Ma, 2022).

However, the efficacy of this transmission mechanism depends on two critical boundary conditions: the internal cognitive capacity of the leadership team and the external technological infrastructure.

First, while ~~a individual~~ CEO's awareness initiates the green path, complex corporate strategic shifts are determined by the collective capability of the top management team (TMT). TMT educational background serves as an ideal proxy for this capacity. A higher collective educational level signifies stronger information-processing capabilities and a broader knowledge structure (Kang et al., 2025). This intellectual capital helps the team better absorb green fund resources and manage the risks of green innovation.

Second, the execution of biased technological progress relies heavily on advanced technological enabling tools. The application of cutting-edge digital technologies and Artificial Intelligence (AI) optimizes production efficiency and energy dispatch systems in real-time (Wu et al., 2024). Corporate digitalization acts as a technological catalyst, accelerating the integration of green capital into actual production processes and maximizing resource utilization efficiency. Therefore, we propose the following two moderation hypotheses:

Hypothesis 4. (H4 / Moderation 1): The average educational level of the Top Management Team (TMT) positively moderates the carbon reduction effect of green funds.

Hypothesis 5. (H5 / Moderation 2): Corporate digital transformation (including AI application) positively moderates the carbon reduction effect of green funds.

3.4 Logical framework of multiple transmission mechanisms

To ensure logical coherence, these five hypotheses do not operate merely in parallel; instead, they form an interconnected, hierarchical transmission framework driven by green funds across three distinct layers.

Specifically, the alleviation of financing constraints serves as the foundational resource layer, providing the vital financial cushion necessary for long-term corporate sustainability and establishing the baseline ~~baseline~~ effect of carbon emission reduction. Built upon this resource foundation, the intermediate governance layer is activated, wherein corporate reputation acts as an external market discipline and internal incentive, while ESG responsibility and CEO environmental awareness function as standardized internal and external driving mechanisms. Driven by these dual reputational and governance pressures, the firm reaches the terminal execution layer, where the allocated capital is translated into green technological innovation and biased technological progress, ultimately reducing micro-level corporate carbon emission intensity.

Furthermore, the operational boundaries of this entire framework are dynamically shaped by critical contextual factors. While the foundational resource injection and the intermediate governance pressures provide the necessary momentum, the final translation into advanced technical execution is bounded internally by management's collective cognitive capacity and externally by technological enabling capabilities. Specifically, a highly educated top management team strengthens joint decision-making efficiency and risk management during environmental transitions, while an advanced corporate digital infrastructure accelerates factor optimization and process efficiency. Together, these boundary conditions serve as catalytic links that bridge individual managerial awareness with collective corporate action, ensuring that green fund capital is effectively mobilized to achieve optimal carbon performance.

4 Measurement models and data description

This study primarily examines the impact of green funds on corporate carbon emissions. The corporate sample comprises Chinese A-share companies from 2010 to 2022, with firm-level data sourced from the CSMAR database. The analysis excludes companies designated as ST or ST*, financial institutions, and service sector enterprises, retaining only those in manufacturing, mining, and water, electricity, and natural gas supply industries. Following preliminary data processing, the final sample comprises 3,299 firms with 13 years of data.

4.1 Benchmark regression model

Q12 The benchmark regression constructed in this paper is shown in Q13 Equation 1:

$$CEI_{jt} = \beta Greenshare_{jt} + \lambda Ctrls_{jt} + \mu_i + I_i \times \tau_t + \varepsilon_{it} \quad (1)$$

Here, CEI_{jt} serves as the dependent variable, while $Greenshare_{jt}$ is the primary explanatory variable of ~~interest.~~, The coefficient β captures the central effect under investigation. $Ctrls_{jt}$ represents a set of control variables, with λ denoting their associated coefficients. μ_i accounts for individual fixed effects, and $I_i \times \tau_t$ incorporates industry-by-time fixed effects. ε_{it} is the idiosyncratic error term. A detailed description of the variables is provided below:

4.1.1 Dependent variable: carbon emission intensity (CEI_{jt})

Following the methodology of Clarkson et al. (2011), this study employs the ratio of corporate carbon emissions to operating revenue as a measure of Carbon Emission Intensity. Drawing upon the approach of Li et al. (2025), energy consumption data for listed enterprises was compiled, with carbon emissions (CE_{jt}) for each firm subsequently calculated using conversion factors sourced from the China Energy Statistical Yearbook (2022). Detailed conversion factors are presented in Li et al. (2025).

The calculation of corporate carbon emissions performance is as follows:

$$CEI_{jt} = \frac{CE_{jt}}{Revenue_{it}} \quad (2)$$

In Equation 2, CE_{jt} denotes a firm's carbon emissions in year t, while $Revenue_{it}$ represents operating revenue, with revenue data retrieved from the CSMAR database.

It is worth noting that while the quality of environmental and energy consumption disclosures among Chinese listed companies varies, this study implements several rigorous procedures to mitigate potential data measurement errors and ensure the reliability of our conclusions. First, during the sample selection process, we strictly exclude firms with severe missing values, continuous zero reports, or logical inconsistencies in their energy consumption data. Second, the energy-to-carbon conversion coefficients utilized are derived from the official China Energy Statistical Yearbook, which provides a standardized and authoritative benchmark that minimizes subjective disclosure bias across firms. Third, to address potential concerns that varying disclosure quality might bias our empirical results, we conduct series of robustness checks in the subsequent analysis, including estimating the baseline model using a trimmed sample (excluding extreme outliers in energy data) and applying alternative proxy variables for carbon performance. The results remain highly consistent, demonstrating that the potential variations in voluntary energy disclosure quality do not systematically distort the core empirical findings of this paper. To further enhance transparency regarding the sample construction process, we provide the following detailed account of sample screening. The initial sample comprises all Chinese A-share listed firms available in the CSMAR database from 2010 to 2022, totalling approximately 35,000 firm-year observations. We then apply the following sequential exclusion criteria: (1) firms designated as ST or ST* are removed (approximately 1,200 observations) due to abnormal financial conditions; (2) financial institutions (banking, securities, insurance, and other financial sectors) are excluded (approximately 2,800 observations) owing to their fundamentally different asset structures and regulatory frameworks; (3) firms with missing energy consumption data required for carbon emission calculation are excluded (approximately 8,500 observations), which constitutes the largest source of sample attrition; (4) observations with consecutive zero-reporting of energy consumption or logically inconsistent energy data are removed (approximately 1,900 observations). The final analytical sample comprises 17,943 firm-year observations covering 3,299 unique firms. We acknowledge that the exclusion of firms with missing energy data may introduce sample selection bias, as firms voluntarily disclosing energy consumption data likely differ systematically from non-disclosing firms in terms of environmental governance awareness, firm size, and regulatory exposure. This limitation is inherent in studies relying on voluntary environmental disclosures in the Chinese market context. Nevertheless, we note that the proportion of excluded firms is broadly consistent with prior studies using similar data sources (e.g., Li et al., 2025), and our robustness checks—including alternative dependent variable measurements and trimmed samples—yield qualitatively identical results, providing confidence that potential selection bias does not fundamentally distort our core findings.

4.1.2 Core variable: green fund shareholding ratio ($Greenshare_{jt}$)

Following ~~Maetal (2023)~~, green public funds are first identified based on fund names, investment objectives, and investment scope, using keywords such as "low carbon," "energy saving," "environmental protection," "carbon neutrality," "emission reduction," "green," "ecology," "new energy," "clean energy," and "natural resources." Subsequently, investment data of these green public funds, including semiannual and annual holdings, are obtained from the CSMAR database. Fund holdings are then matched with fund shareholding disclosures to compile detailed records of green fund investments during each sample period. Finally, these data are matched with listed firms to calculate their green fund shareholding ratios. Specifically, the aggregate green fund holdings in a firm for a given year are divided by the total tradable shares of the firm, yielding the proportion of shares held by green funds at the firm level (%).

4.1.3 Control variables ($Ctrls_{jt}$)

This study incorporates a series of control variables to account for firm-level heterogeneity. Specifically, return on equity (ROE_{jt}) is measured as net profit relative to shareholders' equity. Leverage (Lev_{jt}) is defined as the ratio of total liabilities to total assets. Board size ($\ln Board_{jt}$) is represented by the logarithm of the number of directors, while the proportion of independent directors ($Indep_{jt}$) is calculated as the share of independent directors on the board. Ownership concentration ($Top5_{jt}$) is proxied by the percentage of shares held by the top five shareholders. Auditor quality ($Big4_{jt}$)

is a binary indicator equal to 1 if the firm engages one of the Big Four accounting firms, and 0 otherwise. Cash flow ratio ($CashFlow_{jt}$) is measured as net operating cash flow divided by total assets. Inventory intensity (INV_{jt}) is the ratio of inventory to total assets. CEO duality ($Dual_{jt}$) is a dummy variable equal to 1 if the board chair also serves as CEO, and 0 otherwise. Tobin's Q ($TobinQ_{jt}$) reflects the ratio of market value to total assets. Finally, firm age ($\ln FrmAge_{jt}$) is defined as the logarithm of the number of years since the firm's incorporation.

4.2 Regression model for the mediating mechanism

The mediation mechanism test regression constructed in this paper is shown in Equation 3:

$$Med_{jt} = \beta Greenshare_{jt} + \lambda Ctrls_{jt} + \mu_i + I_i \times \tau_t + \varepsilon_{it} \quad (3)$$

In Equation 3, Med_{jt} denotes the mediating variable. Drawing on the theoretical framework of this study, the mediating mechanisms considered include green innovation, financing constraints, ESG responsibility, corporate reputation, and relative factor input. The details are specified as follows:

4.2.1 ESG responsibility (ESG_{jt})

ESG represents an investment philosophy and assessment framework emphasizing corporate practices in Environmental, Social, and Governance dimensions. This study employs ESG rating data provided by Huazheng (China Securities Index).

4.2.2 Corporate reputation ($Reputation1_{jt}$ and $Reputation2_{jt}$)

Corporate reputation is proxied by media coverage, calculated as the natural logarithm of one plus the total number of positive reports in online sources ($Reputation1_{jt}$) and newspapers ($Reputation2_{jt}$) each year. The data are drawn from the China Economic News Database.

4.3 Regression model for regulatory mechanisms

The moderation mechanism test regression constructed in this paper is shown in Equation 4:

$$CEI_{jt} = \beta Greenshare_{jt} \times Mod_{jt} + \lambda Ctrls_{jt} + \mu_i + I_i \times \tau_t + \varepsilon_{it} \quad (4)$$

In Equation 4, Mod_{jt} serves as the moderating variable. Drawing on the theoretical framework of this study, digital technology, artificial intelligence, and executives' educational background are considered as moderating mechanisms. Details are provided as follows:

4.3.1 Digital technology ($Digital_{jt}$)

Using a text analysis approach, this study measures firms' digital technology adoption by examining disclosures in annual reports related to "digital technology." Keywords include big data, cloud computing, blockchain, artificial intelligence, Internet of Things, 5G, mobile Internet, virtual reality (VR), augmented reality (AR), deep learning, machine learning, digital twin, distributed computing, and edge computing. The frequency of these keywords in annual reports is calculated, and the natural logarithm of one plus the frequency (DT) is used as an indicator of digital technology application.

TABLE 1 Descriptive statistics.

VarName	Obs	Mean	SD	Min	Median	Max
CEI_{jt}	17,943	0.488	0.583	0.043	0.180	2.347
CE_{jt}	20,646	0.029	0.034	0.000	0.018	0.257
$Greenshare_{jt}$	20,646	0.001	0.003	0.000	0.000	0.047
ROE_{jt}	20,646	0.064	0.121	−0.593	0.069	0.383
Lev_{jt}	20,646	0.408	0.190	0.055	0.403	0.876
$\ln Board_{jt}$	20,645	2.128	0.196	1.609	2.197	2.708
$Indep_{jt}$	20,645	0.374	0.052	0.333	0.333	0.571
$Top5_{jt}$	20,646	0.525	0.152	0.197	0.523	0.885
$Big4_{jt}$	20,646	0.049	0.216	0.000	0.000	1.000
$CashFlow_{jt}$	20,646	0.052	0.065	−0.135	0.050	0.242
INV_{jt}	20,646	0.130	0.089	0.000	0.115	0.458
$Dual_{jt}$	20,646	0.286	0.452	0.000	0.000	1.000
$TobinQ_{jt}$	20,646	2.040	1.222	0.864	1.659	8.510
$\ln FrmAge_{jt}$	20,646	2.876	0.325	1.792	2.890	3.526
ESG_{jt}	20,646	4.084	1.038	1.000	4.000	8.000
$Reputation1_{jt}$	20,646	2.701	1.797	0.000	2.398	8.729
$Reputation2_{jt}$	20,646	4.149	0.998	1.792	4.111	6.982
AI_{jt}	20,646	0.257	0.633	0.000	0.000	3.091
$Digital_{jt}$	20,646	1.240	1.356	0.000	0.693	5.017
$Education_{jt}$	20,646	3.363	0.498	2.000	3.391	5.000

4.3.2 Artificial intelligence (AI_{jt})

Similarly, text analysis is employed to capture the extent of AI adoption by analyzing information related to "artificial intelligence" disclosed in annual reports of listed firms. The keyword list includes terms such as "artificial intelligence," "machine learning," "Internet of Things," and "cloud computing," along with 52 seed words in total. The frequency of these terms in annual reports is counted, and the natural logarithm of one plus the frequency (AI) is used to measure the level of AI application.

4.3.3 Educational background of managers ($Education_{jt}$)

The educational backgrounds of senior executives such as board members and supervisors in listed companies were categorised as follows: 1 = secondary technical school or below, 2 = college diploma, 3 = bachelor's degree, 4 = master's degree, 5 = doctoral

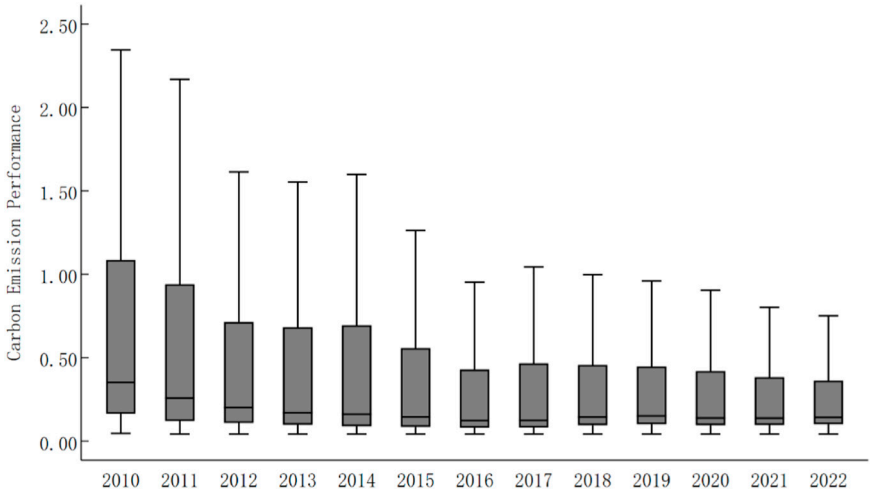


FIGURE 1
Box plot of Carbon Emission Intensity over time. Note: In the box plot, the solid line through the centre of the box represents the annual mean. The solid lines at the top and bottom of the box denote the upper and lower quartiles respectively. The horizontal lines extending from the top and bottom of the solid line through the box represent the adjusted maximum and minimum values. The box plot does not display outliers in the data.

degree. The mean value was then calculated to represent the average educational attainment of management personnel. Data were sourced from the CSMAR database. Descriptive statistics for all variables in this paper are presented in Table 1.

It is worth noting that the median value of green fund holdings (Greenshare) is zero, indicating that more than half of the sampled firms have no green fund holdings during the sample period. This reflects a right-skewed distribution with a large proportion of zero observations. While this distributional feature is noted, we maintain the use of ordinary least squares (OLS) estimation for our baseline regression (Equation 1) because the core independent variable is specified as a continuous proportion. Defining the variable in its original percentage/proportion form allows for a direct and economically meaningful interpretation of the marginal effects—specifically, how a percentage-point change in green fund holdings directly affects corporate carbon emission intensity. Furthermore, to ensure the validity of the OLS estimation under this distribution, our baseline model strictly controls for both firm fixed effects (μ_j) and year fixed effects (ν_t), with standard errors clustered at the firm level. This rigorous specification effectively accounts for unobserved firm-level heterogeneity and time trends, mitigating potential estimation bias and ensuring the robustness of our baseline coefficients.

5 Characteristic facts

5.1 Spatio-temporal analysis of carbon emission intensity

Figure 1 illustrates the temporal evolution of Carbon Emission Intensity among listed Chinese enterprises from 2010 to 2022. The chart reveals that both the median and upper quartile Carbon Emission Intensity exhibited an overall downward trajectory. Notably, the upper quartile demonstrated a more pronounced decline, whereas the lower quartile showed a more gradual shift. This indicates that enterprises with higher carbon emission levels achieved more effective control during this period.

Specifically, based on the sample data in this paper, the average carbon emissions per unit of revenue for Chinese listed companies decreased from 0.67 in 2010 to 0.38 in 2022, while the upper quartile value fell from 1.09 in 2010 to 0.36 in 2022, representing a reduction of over two-thirds. Notably, around the pivotal junctures of 2011 and 2015 – coinciding with the release of China's Climate Change Policy and Action (2011) and the Overall Plan for Ecological Civilisation System Reform respectively–the upper quartile exhibited distinct discontinuous declines. This may indicate the significant constraining effect of government environmental policies on high-carbon-emitting enterprises.

Overall, Figure 1 demonstrates that Chinese enterprises have achieved certain results in carbon reduction, particularly among high-polluting enterprises with poorer Carbon Emission Intensity, where carbon emission intensity has decreased significantly. Conversely, enterprises with inherently low carbon emissions have maintained relatively stable emission performance during this period. This indicates that both enterprises' own green transitions and government policy guidance through green finance, environmental regulations and other means are jointly driving a sustained decline in carbon emissions per unit of output. This provides robust support for the pathway towards achieving the dual carbon goals.

Figures 2, 3 respectively illustrate the spatial distribution patterns of total carbon emissions and unit Carbon Emission Intensity for listed Chinese enterprises in 2010 and 2022. The maps depict carbon emission hotspots at the provincial (autonomous region, municipality) level, aiding in the identification of regional disparities in carbon emissions and their evolving trends.

As observed in Figure 2, the total carbon emissions of China's listed enterprises in 2010 were predominantly concentrated in the central and western regions, with representative provinces including

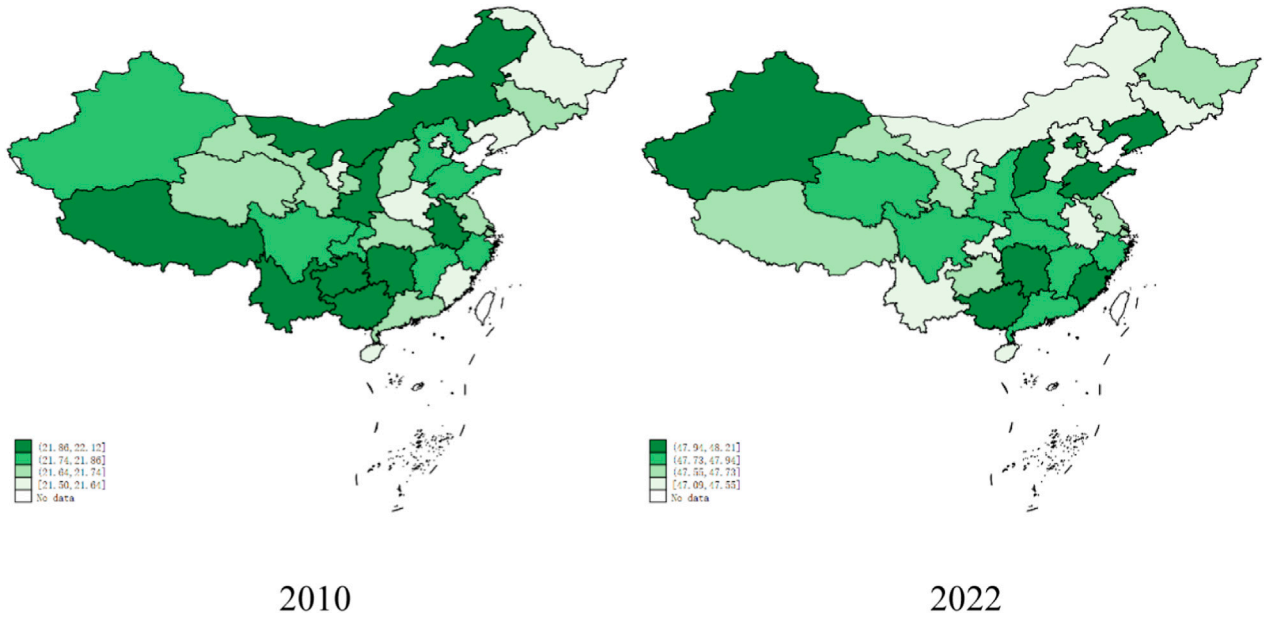


FIGURE 2
Regional heatmap of total carbon emissions from listed Chinese enterprises in 2010 and 2022 (unit: 10,000 tonnes).

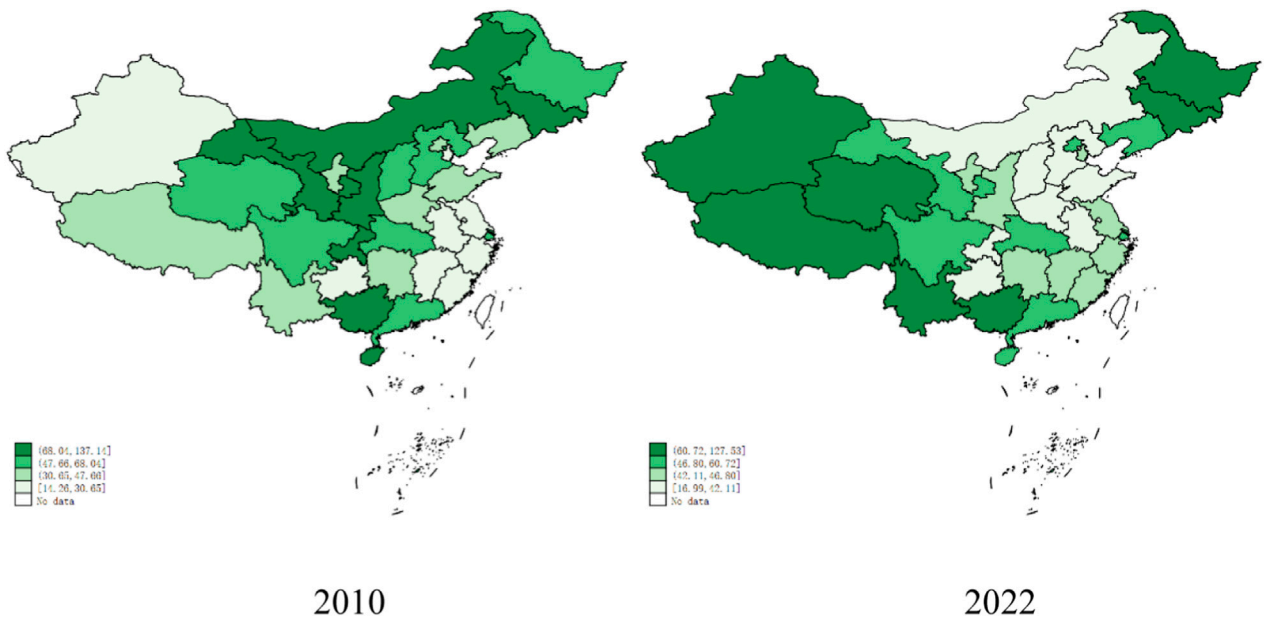


FIGURE 3
Regional heatmap of average carbon emission intensity for listed Chinese enterprises in 2010 and 2022 (unit: Kilograms per Yuan).

Inner Mongolia, Shaanxi, Shanxi, and Yunnan. The high carbon emissions in these regions are closely linked to their resource endowment structure. During their initial development phase, these provinces concentrated a significant volume of high-carbon-intensive industries such as coal, power generation, and non-ferrous metals. The high proportion of coal in their energy mix resulted in elevated carbon emissions per unit of output.

However, by 2022, the regional distribution of total carbon emissions had undergone certain changes. On the one hand, Xinjiang saw a significant rise in its total carbon emissions, while eastern coastal provinces such as Shandong, Fujian and Liaoning exhibited a trend of increasing carbon emission intensity. On the other hand, the growth rate of total emissions slowed in some traditionally high-emission provinces, including Yunnan and Inner

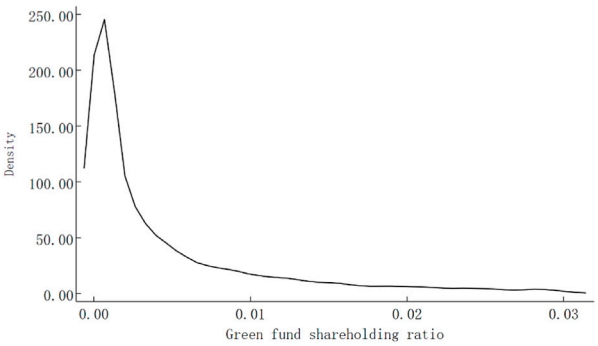


FIGURE 4
Kernel Density Distribution of Green Fund Shareholding Ratios in Chinese Listed Companies. Note: The figure has been trimmed to remove the top and bottom 1% of outliers.

Mongolia, indicating the initial effectiveness of industrial upgrading and energy-saving and emission-reduction policies in certain areas.

As illustrated in Figure 3, the Carbon Emission Intensity per unit indicates that most provinces in the central and western regions exhibited generally poor Carbon Emission Intensity levels in 2010. This reflects their industrial structure being heavily weighted towards energy-intensive sectors and low energy utilisation efficiency. By contrast, regional Carbon Emission Intensity in 2022 exhibited a polarised trend: while some eastern coastal areas maintained high total emissions, their unit Carbon Emission Intensity improved significantly driven by rising shares of high-tech industries and services. Conversely, certain western regions such as Xinjiang, Tibet, and Qinghai—despite lower aggregate emissions—saw their unit Carbon Emission Intensity decline rather than improve, owing to relatively fewer enterprises.

In summary, the evolution of carbon emissions at the regional level reflects that China's green and low-carbon transition is advancing at varying speeds across different regions under the dual carbon goals. The eastern regions exhibit a structural characteristic of 'high total emissions but low carbon intensity', while the western regions face the constraint dilemma of 'low total emissions but high intensity'. This underscores the importance of designing differentiated regional policies in the future.

5.2 Analysis of green fund distribution

Figure 4 illustrates the kernel density distribution of green fund holdings among listed Chinese enterprises. The chart reveals a pronounced bimodal distribution with heavy tails, exhibiting significant right skewness. This indicates that the vast majority of companies maintain low levels of green fund ownership. Statistical analysis of the sample data shows a mean green fund holding ratio of 0.1% and a median of 0, signifying that over half of the enterprises have yet to attract green fund investment. Further analysis reveals that in 2010, approximately 95% of enterprises had no green fund holdings; even by 2022, only 40.3% of enterprises showed records of green fund holdings, indicating that green fund investment coverage in China remains relatively low.

This phenomenon reflects that green funds are still in their nascent stage within China's capital markets, with their investment

TABLE 2 Benchmark test of green fund impact on carbon emission intensity. Q16

	(1)	(2)	(3)	(4)
	CEI_{jt}	CEI_{jt}	CEI_{jt}	CEI_{jt}
$Greenshare_{jt}$	−0.243***	−0.833***	−0.178***	−0.199***
	(-5.07)	(-9.93)	(-4.07)	(-4.55)
ROE_{jt}		−0.054***	−0.020***	−0.020***
		(-19.36)	(-19.46)	(-18.33)
Lev_{jt}		−0.064***	−0.026***	−0.023***
		(-22.72)	(-19.95)	(-18.21)
$\ln Board_{jt}$		−0.021***	−0.007***	−0.006***
		(-9.34)	(-6.55)	(-6.45)
$Indep_{jt}$		−0.010	−0.001	−0.002
		(-1.25)	(-0.51)	(-0.62)
$Top5_{jt}$		0.007**	0.010***	0.006***
		(2.57)	(6.07)	(3.93)
$Big4_{jt}$		−0.011***	−0.002**	−0.001*
		(-8.86)	(-2.13)	(-1.75)
$CashFlow_{jt}$		−0.086***	−0.009***	−0.010***
		(-14.71)	(-6.15)	(-6.98)
INV_{jt}		−0.024***	−0.014***	−0.011***
		(-5.06)	(-5.46)	(-4.43)
$Dual_{jt}$		0.006***	−0.000	−0.000
		(6.74)	(-0.02)	(-0.27)
$TobinQ_{jt}$		0.007***	0.001***	0.001***
		(12.18)	(8.29)	(7.85)
$\ln FrmAge_{jt}$		0.002	−0.008***	−0.019***
		(1.43)	(-12.06)	(-8.78)
_cons	0.029***	0.092***	0.073***	0.103***
	(815.44)	(10.83)	(18.54)	(14.82)
id	Yes	No	Yes	Yes
industry#year	Yes	No	No	Yes
N	20,332	20,645	20,354	20,331
r^2	0.952	0.306	0.954	0.959

*, **, *** denote significance at the 1%, 5%, and 10% levels respectively; t-statistics in parentheses are based on robust standard errors clustered at the enterprise level; the same applies to the table below.

activities yet to permeate comprehensively across the majority of listed enterprises. Particularly following the 2015 release of the 'Overall Plan for Ecological Civilisation System Reform', while green fund investments have seen some growth, the overall progress remains sluggish, failing to achieve widespread coverage or deep engagement. These findings also suggest that in driving corporate green transformation, green funds still possess

TABLE 3 Robustness test of green fund impact on carbon emission intensity.

	(1)	(2)	(3)	(4)	(5)	(6)
	Modify fixed effects		Modify the standard error for clustering	Exclude samples from 2019 onwards	Replace the dependent variable	Tail reduction of 5%
	CEI_{jt}	CEI_{jt}	CEI_{jt}	CEI_{jt}	CEI_{jt}	CEI_{jt}
$Greenshare_{jt}$	−0.981***	−0.192***	−0.199***	−0.126***	−0.411**	−0.199***
	(-11.62)	(-4.40)	(-4.86)	(-4.02)	(-2.16)	(-7.38)
ROE_{jt}	−0.052***	−0.020***	−0.020***	−0.018***	−0.142***	−0.020***
	(-19.37)	(-18.51)	(-15.38)	(-25.30)	(-31.74)	(-31.75)
Lev_{jt}	−0.062***	−0.023***	−0.023***	−0.024***	0.042***	−0.023***
	(-20.69)	(-18.19)	(-14.83)	(-30.10)	(8.25)	(-33.45)
$\ln Board_{jt}$	−0.017***	−0.006***	−0.006***	−0.005***	−0.005	−0.006***
	(-7.89)	(-6.16)	(-6.38)	(-7.03)	(-0.94)	(-8.71)
$Indep_{jt}$	−0.007	−0.001	−0.002	−0.000	0.014	−0.002
	(-0.91)	(-0.53)	(-0.63)	(-0.15)	(0.93)	(-0.78)
$Top5_{jt}$	0.005**	0.006***	0.006***	0.005***	0.002	0.006***
	(1.97)	(3.95)	(3.80)	(4.49)	(0.25)	(6.22)
$Big4_{jt}$	−0.010***	−0.001*	−0.001	−0.000	−0.007	−0.001*
	(-8.26)	(-1.65)	(-1.59)	(-0.22)	(-1.45)	(-1.79)
$CashFlow_{jt}$	−0.082***	−0.010***	−0.010***	−0.010***	−0.096***	−0.010***
	(-13.83)	(-6.61)	(-7.08)	(-7.86)	(-11.21)	(-8.71)
INV_{jt}	−0.018***	−0.012***	−0.011***	−0.006***	0.060***	−0.011***
	(-3.47)	(-4.67)	(-4.43)	(-3.66)	(5.62)	(-7.73)
$Dual_{jt}$	0.005***	−0.000	−0.000	−0.000	−0.004**	−0.000
	(6.05)	(-0.08)	(-0.28)	(-0.81)	(-2.44)	(-0.39)
$TobinQ_{jt}$	0.007***	0.001***	0.001***	0.002***	−0.002***	0.001***
	(12.19)	(7.90)	(7.62)	(16.44)	(-3.92)	(13.38)
$\ln FrmAge_{jt}$	−0.002	−0.018***	−0.019***	−0.017***	0.023**	−0.019***
	(-0.98)	(-8.42)	(-8.10)	(-11.78)	(2.55)	(-15.16)
_cons	0.091***	0.101***	0.103***	0.093***	0.420***	0.103***
	(10.35)	(14.34)	(13.89)	(19.22)	(13.50)	(24.52)
id	No	Yes	Yes	Yes	Yes	Yes
industry	Yes	No	No	No	No	No
year	Yes	No	No	No	No	No
industry#year	No	Yes	Yes	Yes	Yes	Yes
province#year	No	Yes	No	No	No	No
N	20,644	20,330	20,331	14,927	17,625	20,331
r^2	0.351	0.960	0.959	0.961	0.994	0.959

considerable scope for development and potential for policy-driven guidance.

6 Empirical analysis

6.1 Benchmark tests

Table 2 presents the estimated impact of Greenshare on Carbon Emission Intensity (CEI). When no control variables are included, considering only firm-level fixed effects and industry × time fixed effects (Column 1), Greenshare's coefficient is −0.243, statistically significant at the 1% level. This indicates that higher Greenshare participation correlates with lower CEI values, signifying improved Carbon Emission Intensity. After incorporating all firm-level control variables alongside firm-level and industry × year fixed effects (Column 4), the coefficient adjusts slightly to −0.199 (at the 1% level), yet the negative relationship remains robust. These findings align with existing research, confirming the robustness of our analysis and supporting the view that green funds significantly enhance listed companies' Carbon Emission Intensity (Li et al., 2025; Wang et al., 2024).

6.2 Robustness tests

6.2.1 Controlling fixed effects at other levels

To account for high-dimensional unobservable factors potentially influencing corporate Carbon Emission Intensity, we replaced the fixed-effects structure employed in the baseline regression with two alternative specifications. Column (1) of Table 3 reports results where industry and year fixed effects replaced the firm and industry × year fixed effects; Greenshare's coefficient was −0.981 and remained significant at the 1% level. ~~In Column~~ Column (2) adds province × year fixed effects to the original company and industry × year effects, yielding an estimated Greenshare coefficient of −0.192, which remains statistically significant at the 1% level. This finding indicates that our core result—that increased green funding significantly improves corporate Carbon Emission Intensity—remains robust even after controlling for broad industry trends, common temporal shocks, and region-specific time factors.

6.2.2 Modifying the robust standard error for clustering

To comprehensively account for potential correlations at the industry-provincial level, we re-clustered robust standard errors by industry × provincial (rather than firm) level. Column (3) of Table 3 reports the final estimates: Greenshare's coefficient stands at −0.199, remaining statistically significant at the 1% level. This confirms our foundational discovery that increased green fund holdings substantially improve Carbon Emission Intensity, even when accounting for industry-provincial-level error correlation. This further demonstrates the robustness of our findings.

6.2.3 Mitigating the impact of significant events

Certain external shocks (such as the outbreak of the COVID-19 pandemic) may distort our baseline estimates. To rule out this possibility, we re-estimated the model after excluding all firm-year observations from 2020 onwards. Column (4) of Table 3 reports the results: the Greenshare coefficient stands at −0.126, remaining statistically significant at the 1% level. This finding indicates that even when excluding the period most severely impacted by the pandemic, green funds continue to exert a significant positive influence on listed companies' carbon emissions performance. ~~Certain external shocks (such as the outbreak of the COVID-19 pandemic) may distort our baseline estimates. To rule out this possibility, we re-estimated the model after excluding all firm-year observations from 2020 onwards. Column (4) of Table 3 reports the results: the Greenshare coefficient stands at −0.126, remaining statistically significant at the 1% level. This finding indicates that even when excluding the period most severely impacted by the pandemic, green funds continue to exert a significant positive influence on listed companies' carbon emissions performance.~~

6.2.4 Altering the method of measuring the dependent variable

Drawing upon the research findings of Wang et al. (2023), we have developed an alternative metric for Carbon Emission Intensity, denoted as CE. This is calculated by first estimating the total emissions of company j in year t, as follows:

$$CE_{jt} = \left(NC_{it} \times CC \times \frac{Cost_{jt}}{Cost_{it}} \right) \Big/ Revenue_{jt} \tag{5}$$

~~In the equation above~~, CE_{jt} denotes the Carbon Emission Intensity of firm j in year t, NC_{it} refers to the total energy consumption of industry i, while CC represents the carbon conversion factor, with the value for one ton of standard coal set at 2.492. ~~$Cost_{it}$ captures the operating costs of firm j in year t, whereas $Cost_{jt}$ indicates the aggregate operating costs of industry i.~~ $Revenue_{jt}$ measures the operating revenue of firm j in year t.

Table 3, column (5) reports the estimated results for listed companies' Carbon Emission Intensity calculated using an indirect approach. The coefficient for green funds is −0.411, remaining statistically significant at the 1% level. This finding indicates that even under this alternative method of indirectly calculating Carbon Emission Intensity, a higher proportion of green funds can still significantly improve listed companies' Carbon Emission Intensity, thereby reinforcing the robustness of our main conclusions.

6.2.5 Mitigating the impact of outliers

To mitigate potential bias from extreme observations, we applied a 5% tail trimming to the CE variable and re-estimated the baseline regression. Column (6) of Table 3 reports the results: the coefficient for Greenshare is −0.199 and remains statistically significant at the 1% level. This indicates that even after excluding the highest and lowest 5% of Carbon Emission

TABLE 4 Robustness test of addressing potential greenwashing in green fund identification.

Variables	(1) Baseline CEI	(2) Strict green fund definition
Strict_Greenshare		−0.187*** (−3.92)
Greenshare	−0.199*** (−4.55)	
Control variables	Yes	Yes
Firm FE	Yes	Yes
Industry × year FE	Yes	Yes
N	17,943	17,943
R^2	0.158	0.155

*, **, *** denote significance at the 1%, 5%, and 10% levels respectively. Strict_Greenshare is calculated using only funds that meet both keyword and portfolio allocation criteria. t-statistics in parentheses are based on robust standard errors clustered at the enterprise level.

Intensity values, a higher proportion of green fund holdings still significantly improves listed companies' Carbon Emission Intensity, thereby confirming the consistency of our baseline findings.

6.2.6 Addressing potential greenwashing in green fund identification

A potential concern with our green fund identification strategy is that keyword-based matching may misclassify "greenwashing" funds—those that include green-related terms in their names or stated investment objectives but do not genuinely pursue environmentally responsible investment strategies—as legitimate green funds. Such misclassification would introduce measurement error in the key independent variable, potentially attenuating or biasing the estimated coefficients. To address this concern, we adopt two complementary approaches. First, following the methodology of Flammer (2021) and Chi et al. (2023), we construct a "strict" green fund identification criterion that requires funds to not only contain green keywords in their names but also to allocate at least 60% of their portfolio to firms classified as green or environmental under China's "Green Bond Endorsed Project Catalogue" or the "Catalogue for Guiding Industry Restructuring." This stricter definition substantially reduces the likelihood of including greenwashing funds. Second, we exclude funds that were established within the most recent 2 years of each sample period, as newly established funds may be more prone to "theme-chasing" behaviour without substantive green investment track records. Using the strict green fund definition, we re-calculate the green fund shareholding ratio and re-estimate the baseline model. The results, reported in Table 4, show that the coefficient for the strict green fund shareholding ratio remains negative and statistically significant at the 1% level (coefficient = −0.187, t = −3.92), consistent with our baseline findings. This suggests that potential greenwashing in fund identification does not fundamentally alter our conclusions.

6.3 Endogeneity tests

6.3.1 Instrumental variable method test

To address potential endogeneity concerns arising from reverse causality and omitted variable bias, this study employs the instrumental variable (IV) approach with two-stage least squares (2SLS) estimation. Conventional literature frequently utilizes the industry-year mean of green fund holdings as an instrument. However, as noted by peer reviewers, this peer-average approach is prone to violating the exclusion restriction due to industry-wide common shocks, industry norms, and reputation spillover effects. To overcome these limitations and rule out contemporaneous reverse causality, this study selects the one-year lagged inclusion in the MSCI China ESG Leaders Index ($MSCI_ESG_{i,t-1}$) as an exogenous policy shock instrument for green fund holdings ($Greenshare_{i,t}$).

The chosen lagged instrument satisfies both the relevance and exclusion restriction requirements. Regarding relevance, the MSCI China ESG Leaders Index serves as a primary benchmark for passive asset allocation by institutional investors and ESG-themed funds. Inclusion in this index in period $t-1$ triggers mechanical, forced capital flows by index-tracking green funds, thereby significantly increasing the firm's green fund holding ratio in period t ($Greenshare_{i,t}$). Regarding the exclusion restriction, constituent adjustments in period $t-1$ follow standardized global evaluation criteria based on historical disclosures prior to period t. Because the index rebalancing decision predates contemporaneous corporate decisions, $MSCI_ESG_{i,t-1}$ cannot be directly driven by unobserved contemporaneous carbon abatement shocks in period t. Consequently, $MSCI_ESG_{i,t-1}$ influences corporate carbon emission intensity ($Carbon_{i,t}$) exclusively through the delayed channel of green capital allocation, satisfying the exogeneity assumption after controlling for firm and year fixed effects.

The two-stage regression specification is formulated as follows:

$$\text{First Stage: } Greenshare_{i,t} = \alpha_0 + \alpha_1 MSCI_ESG_{i,t-1} + \gamma X_{i,t} + \mu_i + \lambda_t + \varepsilon_{i,t}$$

$$\text{Second Stage: } Carbon_{i,t} = \beta_0 + \beta_1 \widehat{Greenshare}_{i,t} + \delta X_{i,t} + \mu_i + \lambda_t + \eta_{i,t}$$

In the first-stage model, $MSCI_ESG_{i,t-1}$ is a dummy variable equal to 1 if firm i was included in the MSCI China ESG Leaders Index in year $t-1$, and 0 otherwise. In the second-stage model, $\widehat{Greenshare}_{i,t}$ represents the fitted values of green fund holdings obtained from the first stage, and $Carbon_{i,t}$ denotes corporate carbon emission intensity. $X_{i,t}$ represents the set of control variables. μ_i and λ_t capture firm and year fixed effects, respectively.

The diagnostic test statistics presented in Table 4 confirm the statistical validity of the lagged instrument. The Kleibergen-Paap rk LM statistic for the underidentification test is 27.105 ($p < 0.001$), rejecting the null hypothesis of underidentification at the 1% significance level. Furthermore, the Kleibergen-Paap rk Wald F statistic reaches 27.563, which exceeds the strict Stock-Yogo critical value of 26.87 at the 5% maximal IV bias level, alleviating concerns about weak instruments.

Q14 TABLE 5 reports the 2SLS estimation results alongside relevant diagnostic tests.

Variables	(1) First stage greenshare	(2) Second stage carbon emission intensity
MSCI_ESG	0.876*** (5.25)	—
Greenshare	—	−0.138** (−2.52)
Constant	−0.1482*** (−3.71)	0.2730*** (6.21)
Firm FE	Yes	Yes
Year FE	Yes	Yes
Controls	Yes	Yes
Underidentification test (Kleibergen-Paap rk LM)	—	27.105*** [0.000]
Weak IV test (Kleibergen-Paap rk Wald F)	—	27.563 {26.87}
Observations	19,820	19,820
R^2	0.308	0.281

Q17 *p*-value is in square brackets []. The value in curly braces {} indicates the Stock-Yogo weak identification test critical value at the 5% maximal IV bias level.

In the first-stage estimation reported in Column (1), the coefficient of $MSCI_ESG_{i,t-1}$ is positive and statistically significant at the 1% level ($\alpha_1 = 0.876, t = 5.25$), confirming that previous-period index inclusion significantly boosts subsequent green fund holdings. In the second-stage estimation reported in Column (2), the coefficient of fitted green fund holdings ($\widehat{Greenshare}_{i,t}$) on carbon emission intensity ($Carbon_{i,t}$) remains negative and statistically significant at the 5% level ($\beta_1 = -0.138, t = -2.52$). These empirical results demonstrate that after eliminating contemporaneous endogeneity bias through a lagged exogenous index inclusion shock, green fund holdings exert a robust mitigating effect on corporate carbon emission intensity, fully confirming the primary theoretical hypothesis.

6.3.2 Lagged explanatory variable test

To further mitigate the potential endogeneity issues arising from simultaneous causality and measurement errors, this study employs a lagged explanatory variable approach as an additional robustness check. Considering that the implementation of green funds' investment strategies, participation in corporate governance, and the subsequent adjustments in corporate environmental behavior require a certain period to materialize, their environmental governance effects may exhibit a temporal lag. Therefore, we lag the core explanatory variable, green fund shareholding ratio ($Greenshare_{jt}$), by one period ($L.Greenshare_{jt}$) and re-estimate the benchmark model. This specification ensures that the explanatory variable precedes the dependent variable chronologically, effectively eliminating simultaneous reverse causality.

Table 6 presents the empirical results of the lagged explanatory variable test. Column (1) reports the baseline estimation including only the one-period lagged green fund shareholding ratio ($L.Greenshare_{jt}$), firm fixed effects, and industry-by-year fixed effects. The coefficient of $L.Greenshare_{jt}$ is −0.176 and is statistically significant at the 1% level. Column (2) incorporates all firm-level control variables on top of the high-dimensional fixed effects. The coefficient of $L.Greenshare_{jt}$ remains significantly negative at −0.154 ($t = -3.85$), passing the significance test at the 1% level.

TABLE 6 Lagged explanatory variable test results.

Variables	(1) CEI_{jt}	(2) CEI_{jt}
$L.Greenshare_{jt}$	−0.176***	−0.154***
	(-4.12)	(-3.85)
Control variables	No	Yes
Firm FE	Yes	Yes
$Industry \times Year\ FE$	Yes	Yes
N	17,032	17,032
R^2	0.955	0.96

These findings indicate that even after blocking simultaneous feedback channels, a higher proportion of green fund holdings in the previous year continues to exert a significant suppressing effect on the current year's corporate carbon emission intensity. The magnitude and sign of the lagged coefficient are highly consistent with the benchmark regression results in Table 2, confirming that green funds act as a stable driver for corporate low-carbon transformation rather than merely chasing firms with pre-existing superior carbon performance. This further strengthens the causal inference of our baseline conclusions.

6.4 Heterogeneity testing

As a vital financial instrument supporting sustainable development, the effectiveness of green funds in improving corporate Carbon Emission Intensity may vary depending on

firm characteristics and external environments. This study conducts a detailed heterogeneity analysis from four perspectives: industry attributes, regional environmental regulation intensity, corporate life cycle stage, and industry market concentration. Given the differences in model specifications across subgroups, direct comparisons of coefficient magnitudes may lead to biased conclusions. Therefore, we employ the Bootstraps between-groups difference (Bdiff) test proposed by Lian (2016) to examine whether differences in estimated coefficients across subgroups are statistically significant, thereby enabling meaningful interpretation of marginal effects under varying conditions.

6.4.1 Heterogeneity in industry attributes

Environmental policy pressures exhibit significant variations across different industry types. Following the classification criteria issued by China's Ministry of Ecology and Environment in 2008, we categorised the sample into non-heavily polluting industries and heavily polluting industries. The results are presented in columns (1) and (2) of Table 6. We observe that the green fund participation coefficient for non-heavily polluting industries is −0.226, significant at the 1% level, whereas the coefficient for heavily polluting industries is −0.042, statistically insignificant. This indicates that green fund participation exerts a more pronounced effect on improving Carbon Emission Intensity within non-heavily polluting industries.

The fundamental reason may lie in differences in marginal abatement costs and adjustment flexibility across sectors. Non-polluting enterprises typically face lower compliance thresholds and possess more resilient production systems, enabling them to respond more readily to green finance signals through technological upgrades or cleaner inputs. In contrast, heavily polluting industries (often capital-intensive and technologically rigid) may encounter greater institutional and cost frictions in achieving green transition objectives, thereby diminishing the impact of green funds. The pronounced divergence between these two groups further supports the view that the marginal effects of green investment are more pronounced in sectors with greater environmental adaptability and lower transition barriers. Furthermore, the insignificant coefficient for heavily polluting industries may also be attributed to the following factors. First, heavily polluting enterprises are subject to more stringent mandatory environmental regulations (e.g., pollutant discharge permits, mandatory cleaner production audits), which already compel firms to adopt emission reduction measures. In this context, the marginal governance effect of green fund shareholdings—primarily operating through market-based and reputational channels—may be relatively limited compared to the coercive force of regulatory mandates. Second, heavily polluting firms typically possess large-scale, specialised production equipment with high sunk costs and long payback periods, rendering short-term capital injections from green funds insufficient to drive fundamental shifts in production technology. Third, the carbon reduction potential of heavily polluting enterprises, while theoretically greater in absolute terms, may require coordinated policy interventions (e.g., carbon trading, green subsidies) that extend beyond the scope of equity-based green fund governance. These findings suggest that green fund strategies targeting heavily polluting sectors may need to be complemented by stronger regulatory and fiscal policy instruments to achieve meaningful emission reductions.

6.4.2 Regional heterogeneity in environmental regulatory intensity

The intensity of environmental regulation reflects the government's commitment to ecological governance at the regional level. We calculate the ratio of these keywords to the total word count based on their frequency in China's provincial government work reports, using this as a proxy indicator for regional environmental regulation intensity (ER). Regions are classified into high-regulation groups and low-regulation groups according to the median value of ER. The estimated results are presented in columns (1) and (2) of Table 8. Column (1) shows that in regions with weaker regulation, the coefficient for Greenshare is −0.088 and is significant at the 5% level, indicating that participation in green funds moderately enhances firms' Carbon Emission Intensity. By contrast, in regions with stronger regulation, the Greenshare coefficient stands at −0.238 and is statistically significant at the 1% level. This indicates that under greater regulatory pressure, the positive impact of green fund participation on corporate Carbon Emission Intensity is significantly amplified.

Differences in marginal effects may be attributed to varying compliance incentives among enterprises. In regions with stringent environmental regulation, businesses face greater external pressure to adhere to sustainability mandates. Green funds are more likely to be embraced as strategic tools for meeting regulatory expectations by signalling environmental commitment and offering reputational or financial backing. Conversely, in areas with weaker enforcement, enterprises may lack institutional pressure to respond effectively to green capital, thereby diminishing its impact.

6.4.3 Heterogeneity in the corporate life cycle

Due to differences in resource requirements, strategic orientation, and operational flexibility, enterprises at distinct stages of their life cycle may exhibit varying degrees of responsiveness to green fund interventions. Drawing upon the classification methodology proposed by Hribar and Yehuda (2015), we assess enterprises' life cycle stages based on four indicators: sales growth rate, retention rate, capital expenditure ratio, and company age. Following industry-specific scoring and ranking of these variables, enterprises are categorised into three phases: growth, maturity, and decline.

Regression results are presented in Columns (1) to (3) of Table 14. In Column (1), the Greenshare coefficient for growth-phase enterprises is −0.322, significant at the 1% level, indicating a pronounced positive impact of green funds on Carbon Emission Intensity. In column (2), the Greenshare coefficient for mature-stage firms is −0.085 and statistically significant at the 10% level, indicating a relatively weaker yet still positive impact of

TABLE 7 Industry heterogeneity test of green fund impact on carbon emission intensity.

	(1) Non-heavily polluting industries CEI_{jt}	(2) Heavily polluting industries CEI_{jt}
$Greenshare_{jt}$	−0.226***	−0.042
	(-4.45)	(-0.98)
ROE_{jt}	−0.024***	−0.009***
	(-16.79)	(-8.61)
Lev_{jt}	−0.027***	−0.011***
	(-16.81)	(-8.03)
$\ln Board_{jt}$	−0.008***	−0.001
	(-6.07)	(-1.48)
$Indep_{jt}$	−0.003	0.004
	(-0.85)	(1.57)
$Top5_{jt}$	0.011***	−0.000
	(4.78)	(-0.16)
$Big4_{jt}$	−0.001	−0.001
	(-1.30)	(-1.39)
$CashFlow_{jt}$	−0.010***	−0.012***
	(-4.82)	(-7.69)
INV_{jt}	−0.009***	−0.010***
	(-3.14)	(-3.05)
$Dual_{jt}$	−0.000	−0.000
	(-0.16)	(-0.69)
$TobinQ_{jt}$	0.001***	0.002***
	(6.19)	(6.81)
$\ln FrmAge_{jt}$	−0.019***	−0.016***
	(-6.70)	(-5.94)
_cons	0.110***	0.072***
	(12.33)	(8.53)
id	Yes	Yes
industry#year	Yes	Yes
Bootstraps inter-group difference test	0.022**	
N	14,294	6009
r^2	0.957	0.972

Greenshare on Carbon Emission Intensity. In contrast, for companies in the decline stage (column (3)), the coefficient is −0.046 and statistically insignificant, suggesting Greenshare ownership exerts negligible influence on Carbon Emission Intensity at this stage.

Heterogeneity across life cycle stages can be explained by differences in financial flexibility and environmental adaptability. Companies in the growth stage are more likely to face financing constraints, thus possessing stronger incentives to actively engage with green investors to demonstrate environmental responsibility and attract external capital. Mature companies, while more stable,

TABLE 8 Test of environmental regulation heterogeneity on green fund impacts on carbon emission intensity.

	(1) Weak environmental regulation CEI_{jt}	(2) stringent environmental regulations CEI_{jt}
$Greenshare_{jt}$	−0.088**	−0.238***
	(-2.30)	(-3.48)
ROE_{jt}	−0.016***	−0.022***
	(-13.92)	(-13.42)
Lev_{jt}	−0.019***	−0.026***
	(-13.28)	(-13.43)
$\ln Board_{jt}$	−0.004***	−0.009***
	(-3.64)	(-5.54)
$Indep_{jt}$	−0.002	−0.003
	(-0.63)	(-0.72)
$Top5_{jt}$	0.006***	0.008***
	(3.03)	(2.98)
$Big4_{jt}$	−0.001	−0.001
	(-1.11)	(-0.94)
$CashFlow_{jt}$	−0.010***	−0.009***
	(-6.45)	(-4.01)
INV_{jt}	−0.009***	−0.012***
	(-3.28)	(-2.85)
$Dual_{jt}$	−0.001*	−0.000
	(-1.82)	(-0.27)
$TobinQ_{jt}$	0.001***	0.001***
	(6.19)	(4.95)
$\ln FrmAge_{jt}$	−0.020***	−0.018***
	(-8.15)	(-5.21)
_cons	0.099***	0.108***
	(12.26)	(9.47)
id	Yes	Yes
industry#year	Yes	Yes
Bootstraps inter-group difference test	0.044**	
N	9659	9609
r^2	0.975	0.958

may have already optimised most environmental management, leading to diminishing marginal returns. Conversely, companies in the decline stage typically encounter structural and operational inertia, limiting their capacity to respond effectively to green investment pressures.

6.4.4 Heterogeneity in industry concentration

Industry market structure can shape firms' strategic responsiveness to external capital signals. To investigate whether the impact of green fund holdings varies with market concentration, we employ the Herfindahl-Hirschman Index (HHI) to measure

TABLE 9 Testing for heterogeneity in corporate life cycles regarding green fund impact on carbon emission intensity.

	(1) Growth phase CEI_{jt}	(2) Maturity stage CEI_{jt}	(3) Decline phase CEI_{jt}
$Greenshare_{jt}$	−0.322***	−0.085*	−0.046
	(-4.70)	(-1.87)	(-0.86)
ROE_{jt}	−0.018***	−0.013***	−0.017***
	(-11.33)	(-9.01)	(-9.92)
Lev_{jt}	−0.017***	−0.019***	−0.023***
	(-11.08)	(-10.53)	(-11.24)
$\ln Board_{jt}$	−0.007***	−0.004***	−0.004***
	(-5.08)	(-2.82)	(-2.91)
$Indep_{jt}$	−0.001	−0.008*	−0.005
	(-0.14)	(-1.85)	(-1.15)
$Top5_{jt}$	0.010***	0.003	−0.002
	(4.51)	(1.19)	(-0.61)
$Big4_{jt}$	−0.001	0.000	−0.002
	(-1.17)	(0.01)	(-1.31)
$CashFlow_{jt}$	−0.007***	−0.009***	−0.009***
	(-2.90)	(-3.81)	(-3.95)
INV_{jt}	−0.010***	−0.008**	−0.009***
	(-2.60)	(-2.30)	(-2.65)
$Dual_{jt}$	0.001	0.000	0.000
	(1.23)	(0.53)	(0.61)
$TobinQ_{jt}$	0.001***	0.001***	0.002***
	(4.40)	(6.21)	(7.34)
$\ln FrmAge_{jt}$	−0.008**	−0.004	−0.004
	(-2.50)	(-1.12)	(-0.71)
_cons	0.066***	0.056***	0.064***
	(6.58)	(4.85)	(3.23)
id	Yes	Yes	Yes
industry#year	Yes	Yes	Yes
Bootstraps inter-group difference test	0.008***(Growth stage - maturity stage)		
		0.264(Maturity phase - decline phase)	
	0.034**(Growth phase - decline phase)		
N	5839	6276	6349
r^2	0.967	0.970	0.977

TABLE 10 Heterogeneity test of industry concentration affecting carbon emission intensity under green funds.

	(1) Low industry concentration CEI_{jt}	(2) High industry concentration CEI_{jt}
$Greenshare_{jt}$	−0.234***	−0.063*
	(-3.81)	(-1.67)
ROE_{jt}	−0.021***	−0.015***
	(-14.71)	(-9.99)
Lev_{jt}	−0.025***	−0.018***
	(-14.69)	(-11.07)
$\ln Board_{jt}$	−0.006***	−0.004***
	(-5.19)	(-2.99)
$Indep_{jt}$	−0.002	−0.001
	(-0.44)	(-0.43)
$Top5_{jt}$	0.006***	0.005**
	(2.86)	(2.33)
$Big4_{jt}$	−0.001	−0.000
	(-1.31)	(-0.02)
$CashFlow_{jt}$	−0.010***	−0.009***
	(-4.87)	(-4.93)
INV_{jt}	−0.012***	−0.008**
	(-3.78)	(-2.16)
$Dual_{jt}$	−0.001	0.001**
	(-1.64)	(2.11)
$TobinQ_{jt}$	0.001***	0.001***
	(7.22)	(3.91)
$\ln FrmAge_{jt}$	−0.021***	−0.012***
	(-7.15)	(-4.64)
_cons	0.112***	0.076***
	(11.65)	(8.72)
id	Yes	Yes
industry#year	Yes	Yes
Bootstraps inter-group difference test	0.024**	
N	12,396	7563
r^2	0.954	0.976

industry concentration, dividing the sample into high-concentration and low-concentration industries based on the median.

Q18 Columns (1) and (2) of ~~Table 15~~ present the estimation results. In low-concentration industries, the coefficient for Greenshare is −0.234, statistically significant at the 1% level, indicating that green funds significantly enhance firms' Carbon Emission Intensity in more competitive market environments. By contrast, in highly concentrated industries, the coefficient is −0.063, significant only at the 10% level and markedly smaller in magnitude.

This phenomenon may relate to differences in competitive pressures and environmental innovation incentives within industries. In more fragmented sectors, firms face intensified

TABLE 11 Testing the mediating mechanisms of green funds on carbon emission intensity.

VarName	(1) ESG	(2) Reputation1	(3) Reputation2
$Greenshare_{jt}$	8.942***	6.893*	8.898***
	(3.57)	(1.89)	(4.66)
ROE_{jt}	0.332***	0.449***	0.739***
	(4.14)	(5.45)	(13.31)
Lev_{jt}	−0.749***	0.298***	0.425***
	(-8.33)	(2.85)	(6.24)
$ln\,Board_{jt}$	0.157*	0.091	0.110*
	(1.70)	(0.90)	(1.69)
$Indep_{jt}$	1.482***	0.328	0.202
	(5.59)	(1.14)	(1.19)
$Top5_{jt}$	0.254*	0.320**	0.021
	(1.84)	(2.03)	(0.20)
$Big4_{jt}$	0.165**	0.120	0.126**
	(2.01)	(1.25)	(2.15)
$CashFlow_{jt}$	−0.500***	−0.092	−0.072
	(-4.01)	(-0.68)	(-0.81)
INV_{jt}	−0.256	−0.481**	−0.367***
	(-1.31)	(-2.06)	(-2.62)
$Dual_{jt}$	−0.023	−0.012	0.006
	(-0.86)	(-0.41)	(0.35)
$TobinQ_{jt}$	−0.020**	0.064***	0.160***
	(-1.98)	(5.96)	(21.71)
$ln\,FrmAge_{jt}$	−0.174	−0.137	−0.295**
	(-0.97)	(-0.66)	(-2.34)
_cons	3.940***	2.392***	4.169***
	(6.53)	(3.43)	(9.80)
id FE	Yes	Yes	Yes
industry#year FE	Yes	Yes	Yes
N	20,331	20,331	20,331
R^2	0.577	0.815	0.770

market competition and stronger differentiation drivers, making them more inclined to improve environmental performance and cultivate a positive corporate image through green investments. Against this backdrop, green funds can convey credible environmental commitments to stakeholders, thereby strengthening firms' incentives to adopt cleaner production methods. In highly concentrated industries, dominant market players often possess substantial market power and face relatively weaker external constraints, leading to a slower response to green financing signals and a correspondingly diminished marginal incentive effect of green funds.

6.5 Mechanism analysis

6.5.1 ESG responsibility mechanism

Corporate ESG performance is closely correlated with carbon emission efficiency. Higher ESG standards not only demonstrate proactive environmental stewardship and resource utilisation but also reflect comprehensive capabilities in stakeholder relations management and internal governance optimisation. These factors collectively facilitate energy structure optimisation and the implementation of low-carbon production methods (Qian and Liu, 2024; Xie, 2024). Consequently, ESG performance may be regarded as a key transmission mechanism through which green funds influence corporate Carbon Emission Intensity.

As indicated by the regression results in column (1) of Table 11, the coefficient for $Greenshare_{jt}$ in the ESG performance regression is 8.942, and is significantly positive at the 1% significance level. This signifies that an increase in green fund shareholding significantly enhances a company's overall ESG performance. As a significant capital force guided by sustainable development principles, the involvement of green funds may prompt companies to prioritise environmental and social responsibility within their operational strategies. This, in turn, drives improvements in governance structures and overall management standards, leading to higher ESG ratings.

Within the environmental dimension, heightened ESG standards typically entail increased investment in energy-saving technologies, systematic carbon accounting, and stricter enforcement of emission standards (Bratten et al., 2026; Chen et al., 2025). On the social front, closer stakeholder relationships and heightened public scrutiny amplify reputational incentives for emissions reduction while diminishing tolerance for environmentally harmful practices (Bolton and Kacperczyk, 2021; Flammer, 2021; Cao et al., 2023; DesJardine et al., 2021). Within governance, enhanced board oversight and transparent decision-making mechanisms align corporate strategy with long-term sustainability objectives, safeguarding carbon reduction initiatives from short-term profit pressures (Bui et al., 2020; Kavadis and Thomsen, 2022; Orazalin et al., 2023). Through these cumulative effects, green fund holdings elevate ESG performance. This improved ESG standing then serves as an institutional and strategic platform for embedding low-carbon practices into core operations, significantly enhancing corporate Carbon Emission Intensity.

6.5.2 Corporate reputation mechanism

Corporate reputation influences Carbon Emission Intensity through the chain of 'visibility–oversight pressure–behavioural improvement'. Media serve as crucial vehicles for public scrutiny, with newspapers and online platforms amplifying corporate environmental performance (Kathuria, 2007). Heightened public attention translates to stronger societal expectations for emissions reduction and sustainability from enterprises (Giacomini et al., 2020). In China, news coverage and community-level social

TABLE 12 Testing the moderating effect of green funds on carbon emission intensity.

	(1) CEI_{jt}	(2) CEI_{jt}	(3) CEI_{jt}
$Greenshare_{jt}$	−0.092	0.188	−0.159***
	(-1.62)	(0.76)	(-3.09)
$Digital_{jt}$	−0.001***		
	(-7.77)		
$Greenshare_{jt}$*$Digital_{jt}$	−0.068***		
	(-2.94)		
$Education_{jt}$		−0.000	
		(-0.27)	
$Greenshare_{jt}$*$Education_{jt}$		−0.115*	
		(-1.70)	
AI_{jt}			−0.001***
			(-4.02)
$Greenshare_{jt}$*AI_{jt}			−0.146***
			(-2.82)
ROE_{jt}	−0.019***	−0.020***	−0.020***
	(-18.29)	(-18.35)	(-18.39)
Lev_{jt}	−0.023***	−0.023***	−0.023***
	(-18.15)	(-18.23)	(-18.25)
$\ln Board_{jt}$	−0.006***	−0.006***	−0.006***
	(-6.07)	(-6.45)	(-6.33)
$Indep_{jt}$	−0.002	−0.002	−0.002
	(-0.61)	(-0.60)	(-0.65)
$Top5_{jt}$	0.006***	0.006***	0.006***
	(3.89)	(3.96)	(3.84)
$Big4_{jt}$	−0.001*	−0.001*	−0.001*
	(-1.91)	(-1.75)	(-1.72)
$CashFlow_{jt}$	−0.011***	−0.010***	−0.010***
	(-7.26)	(-6.99)	(-7.08)
INV_{jt}	−0.012***	−0.011***	−0.011***
	(-4.70)	(-4.43)	(-4.52)
$Dual_{jt}$	−0.000	−0.000	−0.000
	(-0.20)	(-0.27)	(-0.21)
$TobinQ_{jt}$	0.001***	0.001***	0.001***
	(7.99)	(7.87)	(7.88)
$\ln FrmAge_{jt}$	−0.018***	−0.019***	−0.019***
	(-8.85)	(-8.79)	(-8.87)
_cons	0.103***	0.104***	0.104***
	(15.08)	(14.56)	(14.96)
id	Yes	Yes	Yes
industry#year	Yes	Yes	Yes
N	20,331	20,331	20,331
r^2	0.959	0.959	0.959

(*Continued*)

TABLE 12 Continued

constraints have been demonstrated to effectively drive corporate green innovation and low-carbon initiatives (Zhao et al., 2022). Firms with higher reputations are more likely to attract public scrutiny. The resulting pressure to maintain reputation and potential reputational risks increase the external costs of high-emission behaviour. This strengthens constraints on environmental investments, process improvements, and information disclosure, thereby promoting enhanced Carbon Emission Intensity. We measure corporate reputation using two metrics: the annual number of positive online reports received (Reputation1) and the annual number of newspaper reports (Reputation2).

Columns (2) and (3) of Table 11 present the regression results for these two indicators. $Greenshare_{jt}$ exhibits a coefficient of 6.893 for Reputation1 and 8.898 for Reputation2, significant at the 10% and 1% levels respectively. This indicates that firms holding greater green fund holdings are more likely to receive positive media coverage, whether online or in print. A plausible explanation is that green funds incentivise companies to engage in high-profile environmental initiatives by emphasising sustainability and disclosure, thereby effectively communicating these actions to enhance their public image.

Green funds can indirectly improve Carbon Emission Intensity by elevating corporate reputation. As sustainability-oriented investors, their involvement typically entails stricter environmental disclosure requirements and encourages external communication. This facilitates greater positive media coverage across online and print platforms, thereby boosting public recognition. This enhanced reputation not only intensifies societal oversight of corporate environmental commitments but also attracts policy resource allocation and additional market incentives (Dyck et al., 2019; Bolton and Kacperczyk, 2021; Flammer, 2021; Lu, 2025). Consequently, enterprises are driven to continuously optimise production processes and increase emissions reduction investments, ultimately achieving improved Carbon Emission Intensity.

6.6 Analysis of regulatory mechanisms

Table 12 presents the results of the interaction terms between the proportion of holdings by green funds (Greenshare) and three moderating variables—Digital, education, and AI—on corporate Carbon Emission Intensity (CEI). Column (1) and Column (3)

respectively introduce different moderating variables and their interaction terms to verify the potential mechanisms through which green funds enhance corporate carbon reduction performance via different channels.

First, Column (1) reveals that the interaction term between Digital and Greenshare exhibits a significant negative correlation at the 1% significance level, with a coefficient of −0.068. This result indicates that green fund investments yield more pronounced improvements in Carbon Emission Intensity within enterprises possessing advanced digital technology capabilities. This aligns with the preceding analysis of the 'green technology innovation pathway,' wherein digital technology functions as a clean input, enhancing production efficiency while reducing polluting inputs. Consequently, it amplifies the catalytic effect of green funds on biased technological progress and green innovation.

Secondly, in Column (2), education itself exhibits no significant direct effect on Carbon Emission Intensity. However, its interaction term with Greenshare is significantly negative at the 10% level, with a coefficient of −0.115. This implies that management teams with higher educational attainment, upon receiving green fund investments, demonstrate greater capacity to comprehend and implement strategic planning related to green transition, thereby enhancing carbon reduction outcomes. This finding aligns with the 'governance and reputation pathway' analysis—highly qualified management teams are more likely to proactively respond to external capital's green requirements, strengthen information disclosure and ESG practices, thereby enhancing green transition performance.

Finally, Column (3) indicates that the interaction term between AI and Greenshare is significantly negative at the 1% level, with a coefficient of −0.146, and the absolute value of this coefficient is the largest among the three moderator variables. This indicates that AI development most significantly amplifies the positive impact of green funds on corporate Carbon Emission Intensity. This aligns with the theoretical analysis presented earlier—artificial intelligence can substantially reduce the use of polluting inputs by optimising production processes, enhancing energy efficiency, and enabling real-time carbon emission monitoring, thereby magnifying the emission reduction effects of green fund investments (Wu et al., 2024).

Collectively, the results of the three moderation effects support the theoretical mechanism proposed herein: green fund investments exert a more pronounced promotional effect on corporate carbon reduction performance when underpinned by technological empowerment and enhanced management capabilities. Among these, the development of artificial intelligence exhibits the strongest moderating effect, followed by digital technology, while the moderating effect of managerial educational background is relatively weaker yet remains statistically significant. This finding not only validates the potential of green funds in fostering green technological innovation but also underscores that enterprises should actively pursue digital and intelligent transformation while introducing green capital, and prioritise the role of high-calibre management teams in implementing green strategies.

7 Conclusion

Green finance is increasingly becoming a vital force in driving corporate pollution reduction and carbon mitigation while promoting green technological innovation. This study examines listed Chinese enterprises from 2010 to 2022, systematically analysing the impact of green funds on corporate Carbon Emission Intensity. The primary conclusions are as follows:

First, Chinese enterprises have achieved positive outcomes in carbon reduction overall. Notably, high-polluting enterprises with poor Carbon Emission Intensity demonstrated a significant decline in carbon intensity, while enterprises with inherently low carbon emission levels maintained relatively stable performance throughout the study period.

Second, regionally, eastern China exhibits a structural pattern of 'high total emissions but low carbon intensity', indicating substantial economic scale coupled with relatively efficient carbon output per unit of production. Western regions, conversely, display a constrained pattern of 'low total emissions but high intensity', characterised by limited overall emissions but inefficient carbon output per unit of output.

Thirdly, empirical analysis indicates that higher green fund holdings correlate with superior corporate Carbon Emission Intensity. This conclusion remains robust across multiple stability tests and endogeneity treatments, underscoring green funds' significant role in driving corporate low-carbon transformation.

Fourthly, heterogeneity analysis reveals that green funds exert more pronounced and potent carbon reduction effects in non-heavily polluting industries, regions with stringent environmental regulations, sectors with low concentration, and growth-stage enterprises.

Fifth, mechanism analysis further reveals that green funds enhance corporate Carbon Emission Intensity through multiple pathways, including: fostering green innovation and directional technological progress, alleviating financing constraints, strengthening corporate ESG accountability, and activating reputation mechanisms. Additionally, the educational background of corporate executives, AI development levels, and digital technology capabilities exert significant positive moderating effects within these mechanisms, thereby amplifying the emission reduction efficacy of green funds.

Overall, this study not only supplements empirical evidence on green finance's role in corporate decarbonisation but also illuminates its operational mechanisms and variations under heterogeneous conditions. Nevertheless, several limitations should be acknowledged. First, due to data availability constraints, our carbon emission estimates rely on energy consumption data multiplied by emission coefficients rather than direct firm-level carbon emission measurements. The quality and completeness of voluntary energy disclosures vary across firms, which may introduce measurement error and sample selection bias. Second, our green fund identification relies on keyword matching, which may not perfectly distinguish genuine green funds from those engaging in "greenwashing" behaviour, despite our robustness checks addressing this concern. Third, the sample period (2010–2022) encompasses significant policy changes and external shocks (e.g., COVID-19), and while we conduct robustness tests to mitigate their

impact, the time-varying nature of these confounding factors may not be fully captured.

In terms of policy implications, our heterogeneous findings yield several actionable recommendations. First, given that green funds exert stronger emission reduction effects in non-heavily polluting industries, policymakers should consider designing differentiated green fund incentive schemes that provide additional support (e.g., tax incentives, subsidised green credit lines) for green funds targeting heavily polluting sectors, where market-based mechanisms alone may be insufficient. Second, the significant moderating roles of digital technology and AI suggest that integrating green finance with digital transformation policies could amplify emission reduction outcomes. Regulatory bodies could encourage green fund managers to incorporate digital capability assessments into their investment screening criteria. Third, the finding that green funds are more effective in regions with stringent environmental regulations underscores the importance of maintaining and strengthening environmental regulatory frameworks as a necessary complement to green finance market development. Fourth, regarding the greenwashing concern, regulators should establish more rigorous and standardised green fund certification and disclosure requirements, moving beyond name-based identification to mandate periodic reporting of actual green asset allocation ratios and environmental impact metrics. Future research could expand into cross-market comparisons to examine how institutional contexts shape green fund efficacy, investigate sector-specific emission reduction pathways with more granular industry data, and explore the dynamic evolution of green fund governance effects as China's carbon trading market matures.

Data availability statement

Q19 The raw data supporting the conclusions of this article will be made available by the authors, without undue reservation.

Author contributions

~~LL: .~~ WZ: Data curation, Investigation, Methodology, Validation, Writing – original draft. CZ: Conceptualization, Methodology, Resources, Writing – review and editing. Q20

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